Delphes vs. CMS for HH→ $b\bar{b}\tau\tau$:
where we stand, and one decision to take

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Abstract

The $\tau_h\tau_h$ yield deficit that motivated this work (a factor $\sim$5) is closed: the cutflow agrees with CMS at every stage (1.01) and eight of ten NSBI features agree. The two that do not, $m_{\tau\tau}$ and $\Delta R_{\tau\tau}$, are both on the τ side. The $m_{\tau\tau}$ offset is now traced: it is in the τ four-vectors, fakes are excluded, and the Delphes τ energy response carries a one-sided tail CMS does not have — at gen p_T = 20–30 GeV, one Delphes τ in ten has more than twice the median response, against 1.08 on CMS. A multiplicative median map cannot reshape a distribution, so the current correction cannot close this. That leaves three options, priced in Section 2; the measurements are in Section 3.

1 Status

Stock CMS card plus the patches in `docs/card_patch_validation.md`. All tuning is *downstream*, so samples are never re-made; every map is measured on a private 2024 NanoAODv15 anchor, so "tuned" means "measured on CMS and transferred".

quantity	status	how
b-tag efficiency (b, c, light)	✓ tuned	stochastic re-tag from `Jet.Flavor` at anchor UParT-AK4 Medium
b-jet energy scale	✓ tuned	inverted to unity against `GenJet` (self-anchored)
m_{bb} peak and width	✓ matched	follows
τ_h ID efficiency	✓ tuned	anchor `GenVisTau`→DeepTau v2.5 Medium; includes HPS reco inefficiency
τ_h visible mass	✓ tuned	sampled per jet from anchor per-p_T quantiles
τ_h acceptance, jet–τ overlap	✓ identical	$p_T > 20$, $\lvert\eta\rvert < 2.3$; pair first, then $\Delta R > 0.4$
p_T^{miss} resolution	✓ tuned	Gaussian smearing, 16.5 → 32.6 GeV/component
lepton reco efficiency	✓ tuned	per-p_T SFs (1.57 → 1.22 e; 1.30 → 1.13 μ)
di-τ mass estimator	✓ identical	same covariance-free FastMTT both sides
τ_h energy scale	× tuned	CMS/Delphes response ratio — under-corrects, Section 3.1

Table 1: Object status. Every map is derived on the anchor and applied downstream; none needs a card change.

✓ agree	m_{HH} (the κ_λ observable, across $\kappa_\lambda = 0, 1, 5$); m_{bb} (both peak $\sim$112 GeV); p_T^{HH}; $\Delta\phi_{HH}$; p_T^{H1}, p_T^{H2}; ΔR_{bb} (a
× disagree	$m_{\tau\tau}$ ($\sim$7 GeV high, Delphes broader); $\Delta R_{\tau\tau}$ (excess at 2.2–3.2)

Table 2: NSBI inputs at $\kappa_\lambda = 0$; other κ_λ points behave the same. Yields: gen-matched cutflow 1.01, analysis-level 1.18×.

Known limitations. In three buckets, because "known" was covering different things.

Accepted by design. FastMTT is covariance-free (a fixed p_T^{miss} resolution rather than the per-event covariance), but is run identically on both sides, so it cancels in the comparison and bounds the absolute $m_{\tau\tau}$ resolution rather than the agreement. The card runs without pileup, so jet energy *resolution* is untuned and pileup jets cannot exist; both act on the $b\bar{b}$ side, which agrees, and the second cannot be fixed downstream in any case. p_T^{miss} smearing is isotropic and flat where the real resolution is anisotropic and grows with ΣE_T: this reproduces the width but not its structure, and it remains a candidate for $m_{\tau\tau}$ being marginally *broader* on Delphes — Section 3.1 exonerates p_T^{miss} for the *offset*, not for the width.

A scope bound, not a defect. `tau_mistag` is measured on the b-rich signal anchor, and fake rates are strongly jet-flavour dependent, so it is *not* valid for the fake-dominated backgrounds (QCD, W+jets). This does not limit the present signal study, but the samples cannot support a background-heavy analysis until those are measured separately.

Real defects, not yet fixed. Two in the energy-scale *application*: fake τ_h match neither branch of `escale_factor` and are left uncorrected while still receiving a τ-like mass, and the τ branch matches *all* gen τ (`|pid|==15`) including leptonically decaying ones, whereas the map is derived on the hadronic visible set. Both were measured *not* to be the cause of the $m_{\tau\tau}$ offset (Section 3.1), but both are wrong.

Untested — risks, not accepted limitations. **(a)** The sampled τ_h mass is drawn independently of the τ energy and of the partner leg, whereas the true visible mass correlates with decay mode and hence with prong multiplicity and energy sharing. Size unknown. **(b)** Every map is flat-extrapolated above 300 GeV, the last bin edge, so a 500 GeV jet is given the 200–300 bin's value verbatim. Four reasons this is probably benign, none of them yet checked directly:

- *Population.* Only ~2.8% of b-jets exceed 300 GeV.
- *It is not where κ_λ lives.* The $gg \to HH$ amplitude is $\kappa_\lambda \times (\text{triangle}) + (\text{box})$, and the triangle carries an s-channel Higgs propagator falling with $\hat{s}$. The m_{HH} *shape* difference between κ_λ hypotheses is therefore largest near threshold, $m_{HH} \simeq 250$–400 GeV, where the destructive interference is most modified; the high-m_{HH} tail is box-dominated and comparatively κ_λ-insensitive. b-jets from the discriminating region sit well below 300 GeV. (An earlier version of this note asserted the opposite and used it to argue the extrapolation was dangerous; that was wrong.)
- *Flat is the right asymptote.* The v1 card's b-tag formula, $(p_T > 4) \times (0.904 - 3.53/p_T)$, tends to 0.904, so a constant extrapolation matches the underlying behaviour rather than being arbitrary.
- *The escale has little left to do there.* Table 3 shows the Delphes response converging on CMS by 100–300 GeV (1.035 vs 0.970), so the correction is small and slowly varying where the extrapolation begins.

What would change this is a boosted or high-m_{HH} category, where the extrapolated region stops being a 3% tail and becomes the signal region.

2 The decision

A CMS τ_h is an HPS object built from a narrow set of decay products; a Delphes τ_h is an $R = 0.4$ jet carrying `TauTag`. Mass, energy and direction are therefore wrong at source and we correct the first two rather than fix the representation. Section 3.1 shows the correction cannot succeed as built: a multiplicative per-p_T median cannot depopulate a kinematically forbidden region, and the Delphes response is one-sidedly broader at matched median by 3× at low p_T. That is narrower than "only rebuilding works" — hence three options, not two.

(A) Rebuild the τ_h from particle flow. The card already writes the three `EFlow` collections, per its header, precisely for "τ redefinition". HPS-like: seed from a jet, collect charged hadrons and photons in a narrow signal cone (fixed $\Delta R \simeq 0.1$ or p_T-dependent $3.0/p_T$ bounded at 0.05–0.1), form the standard decay-mode hypotheses, sum signal-cone constituents, isolation annulus as a fake handle. Mass, energy and direction become correct *by construction*; for a pheno paper this removes a fast-simulation limitation instead of documenting one. Weeks, but *not* on the critical path — the multi-day item is running the full 5×10^5-event samples and training the NSBI.

Against: it cannot reproduce DeepTau, so the ID and fake-rate maps remain (they work). It is not obviously the cure for $\Delta R_{\tau\tau}$ — CMS HPS also seeds from AK4 jets, so merging scales are similar. *The caveat we most want checked:* Delphes neutral PF candidates are *calorimeter towers* ($\sim 0.02 \times 0.02$ barrel), not clustered PF objects. Since π^0 content sets both the visible mass and the decay mode, an HPS-like reconstruction here would be approximate in exactly the place that matters — possibly no better than the anchor-sampled mass already in use.

(B) Correct the distribution, not the scale. What is forbidden is a multiplicative *median* map, not every downstream correction. For each gen-matched τ-jet set $p_T^{\text{reco}} = p_T^{\text{gen}} \times r$ with r *drawn* from the anchor's response quantiles — the machinery already used for the visible mass, applied to the energy. Reproduces the CMS response distribution including its tail. About a day. *Against:* only covers the $\sim$96% gen-matched (fakes have no gen τ); p_T^{miss} is built from reco objects, so resampling afterwards breaks the balance (we already smear p_T^{miss} independently, so a difference of degree, but it compounds); the τ energy becomes partly synthetic, which for an unbinned NSBI result resting on realistic simulation is defensible but weaker than a rebuild; and it does nothing for $\Delta R_{\tau\tau}$, decay mode or charge.

(C) Accept it and document. Eight of ten features and the yield already agree. The $m_{\tau\tau}$ residual is $\sim$7 GeV on $\sim$100, and NSBI trains on shapes with a fitted normalisation.

What does *not* decide it. Rebuilding is sometimes said to retire three corrections — visible mass, energy scale, visible-τ reference. Badly weighted: the mass map's residual imperfection is worth 0.11% on $\langle m_{\tau\tau} \rangle$, so retiring it buys almost nothing, and the visible-τ reference is a derivation-side simplification. One of the three matters, and it is the energy scale.

Questions for τ expertise.

1. Is tower-granular neutral reconstruction adequate for a useful decay-mode assignment, or does it degrade π^0 content enough that the rebuilt mass is no better than the anchor-sampled one?
2. Signal cone: fixed 0.1 or p_T-dependent, over the 20–60 GeV range that dominates here?
3. Seeding: jets (as HPS) or direct PF clustering? Would either change the efficiency for collimated pairs, $\Delta R_{\tau\tau} \lesssim 1$?
4. Is the $\Delta R_{\tau\tau}$ pattern — over-efficient above 2.5, possibly under-efficient below 1 — recognisable as a jet-versus-HPS effect?
5. Is option (B) acceptable in a pheno paper, or does resampling a reconstructed quantity undercut the claim?
6. Is a 1.18× analysis-level yield excess acceptable for unbinned NSBI, where training uses shapes and normalisation is fitted?

3 The two open questions

3.1 $m_{\tau\tau}$

$m_{\tau\tau} = m_{\text{vis}}/\sqrt{x_1 x_2}$ factorises, so the *visible* di-τ mass m_{vis} — the two τ four-vectors, no p_T^{miss}, no FastMTT — separates a τ scale error from an x-fit error. The two have disjoint fixes.

It is the four-vectors. m_{vis} is offset by *at least* as much as $m_{\tau\tau}$ (peak ratios ≈ 1.24 vs ≈ 1.17), so FastMTT is not the source; if anything it compresses. Both legs are individually harder. With an applied scale of $\simeq 0.85$ and a residual $\simeq 1.24$, the raw Delphes τ are $\sim 46\%$ harder than CMS while the map corrects 15%.

Not fakes. Splitting gen-matched from fake gives fake fractions 4.1% Delphes vs 4.3% CMS with indistinguishable shapes; the gen-matched component carries everything — the peak displacement and an unphysical tail at $m_{\text{vis}} > 125$ GeV, kinematically impossible for $H \to \tau\tau$ and absent on CMS. Same pattern as $\Delta R_{\tau\tau}$.

The response distribution. Plotting the response (reco τ p_T / gen-visible-τ p_T) as a *distribution*, each side divided by its own median so the scale is removed by construction, leaves only what the map cannot change. One control first: without a τ ID, Delphes matches a gen τ to the nearest jet of *any* kind (so a lost τ-jet can match a b-jet), while CMS matches within the `Tau` collection — and mis-matching mimics the same p_T dependence. With the ID required on both sides the tail survives (Table 3, Figure 1):

gen p_T [GeV]	median		q_{95}/median		ratio
	Delphes	CMS	Delphes	CMS	
20–30	1.184	0.989	3.378	1.135	3.0×
30–50	1.114	0.987	1.873	1.105	1.7×
50–100	1.066	0.978	1.325	1.102	1.2×
100–300	1.035	0.970	1.195	1.115	1.07×

Table 3: τ_h energy response, τ ID required both sides. q_{95}/median is pure *shape*: the scale is divided out.

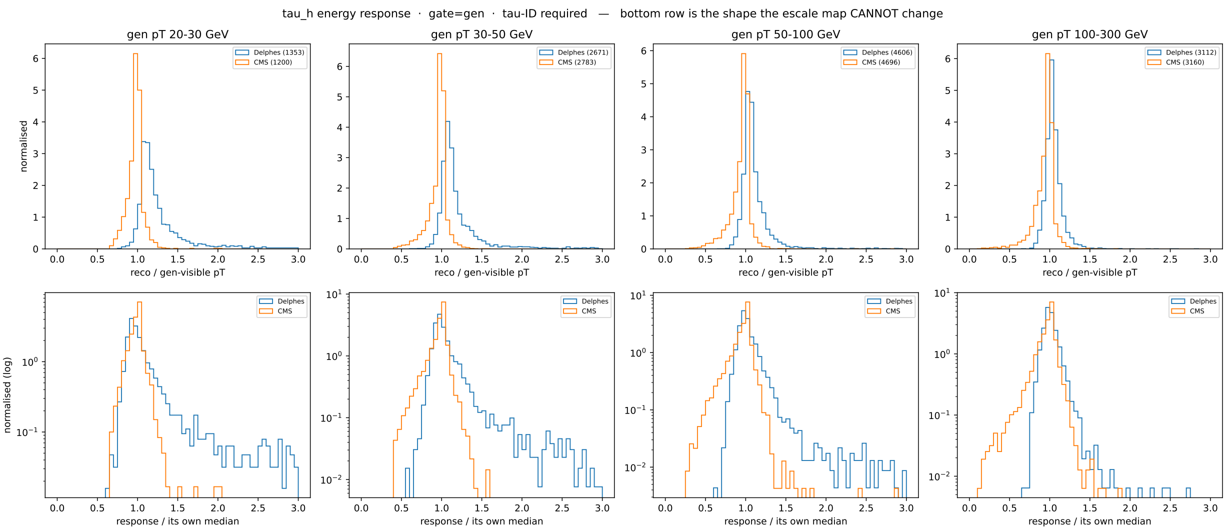

Figure 1: τ_h energy response per gen-p_T bin, τ ID required on both sides. **Top:** as measured — the offset `tau_escale` is built to remove. **Bottom (log):** each side divided by its *own* median, so the scale is gone by construction and only the shape remains — i.e. only what a multiplicative map *cannot* change. The Delphes shelf above 1.5× shrinks with p_T and is consistent with CMS by 100–300 GeV; CMS is flat throughout. Numbers in Table 3.

CMS is flat in p_T (1.10–1.14), as a calibrated object with constant fractional resolution should be (q_{90}/median $= 1.076 \Rightarrow \sigma \simeq 6\%$, right for HPS), so the anchor is sound. Delphes is not flat and converges on CMS as p_T rises — what a constant *absolute* energy excess gives, large as a fraction at 25 GeV and small at 175. Medians converge alongside, as one cause predicts. Sharpest number: at 20–30 GeV the Delphes q_{90}/median is 2.156 against 1.076 on CMS. That is not a resolution. One caveat survives: `TauTag` is a stochastic draw uncorrelated with contamination while DeepTau actively rejects it, so part of the gap is a *selection* correlation — equally unreachable by a scale map, so it does not change Section 2, but the tail is not purely about cone size.

Excluded. *(i) The sampled mass.* The FastMTT floor $x_{\min} = (m_{\text{vis}}^{\text{leg}}/m_\tau)^2$ exceeds a typical fitted $x \simeq 0.5$–0.9 only above ~ 1.26 GeV, so below that the mass is inert: displacing the whole low-mass region by 200 MeV moves $\langle m_{\tau\tau} \rangle$ by 0.00%, against -2.4% in the a_1 tail. Replacing the 21-quantile grid with an atom at $m_{\pi^\pm}$ plus a histogram CDF changes it by 0.11%. *(ii) A symmetric resolution mismatch.* 7% → 25% at fixed median moves the mean 0.05% (RMS +2%) — but that test used a symmetric log-normal, so it does *not* cover the one-sided tail above.

3.2 $\Delta R_{\tau\tau}$: a selection effect, cause unknown

Gen-level $\Delta R_{\tau\tau}$ spectra agree, so the samples are the same physics; the efficiencies differ — Delphes selects 0.22–0.23 at gen $\Delta R > 2.5$ where CMS selects 0.15–0.17 ($\sim 3\sigma$), with a weaker ($\sim 2\sigma$) hint of loss below $\Delta R < 1$. Both push the distribution rightward. Excluded by measurement: fakes (the discrepancy is entirely gen-matched); τ-pair selection (symmetrising moved $\Delta R_{\tau\tau}$, not $m_{\tau\tau}$); CMS τ miscalibration (anchor response 0.97–0.99). This matters more than its size: it is a *selection* effect, so no downstream correction reaches it, and unlike $m_{\tau\tau}$ we have no candidate mechanism.

A Reference

All reproducible from `delphes-pipeline`.

1. **Cutflow** (`tauh_cutflow.py`). Five stages — gen visible τ_h, reco, ID, cleaned jets, FastMTT — walked identically both sides on *conditional* efficiencies. FastMTT once carried the whole deficit (12% vs 99%); now 1.01 overall. Tests inject a loss at one stage and assert it is attributed there only.

2. **FastMTT, and why the mass is sampled.** The hadronic prior vanishes unless $x_{\min} \leq 1$; an AK4 jet mass (5–20 GeV vs $m_\tau = 1.777$) admits no solution, returning NaN and silently removing $\sim$89% of pairs. The anchor mass fixes it, but must be *sampled*: it is a one-sided floor, so a single median relaxes a real constraint for every leg above it, inflating the width $\sim$60% and mean $\sim$9% while leaving the *mode* unchanged — a mode-gated $Z \to \tau\tau$ candle would miss it.

3. **The reference error, twice.** A `GenJet` is also a jet and also contains UE, so it is not the analogue of `GenVisTau`. Against a true visible τ ($p_\tau - p_{\nu_\tau}$, exact for one neutrino) the Delphes τ-jet is 17% harder at 25 GeV, $\sim$2% at 250. The same error appeared in the escale map and in the cutflow's stage 1.

4. **Order of application.** Maps measured at CMS scale must be applied at CMS scale; tagging raw jets evaluated a steep efficiency $\sim$17% too far right. Applying the scale first moved the excess 1.25 → 1.18.

5. **Gen-τ check** (`gen_tau_check.py`). Our visible-τ construction vs CMS `GenVisTau` on the same events: efficiency 0.991, purity 0.998, stage-1 rate to $\sim$1.6%.

6. m_{vis} **and the split** (`nsbi_overlay.py --diagnostics --split-gen-matched`). m_{vis} uses neither p_T^{miss} nor the fit, so it attributes an offset to one of two disjoint causes; tests pin $m_{\text{vis}} \leq m_{\tau\tau}$, invariance under p_T^{miss} 40 → 200 GeV, and exact linearity in the τ scale. The split normalises to each side's total, so the fake curve's area is the fake fraction.

7. **Response shape** (`tau_response.py`). Distribution per gen-p_T bin, lower row divided by each side's own median. Tests assert the separation: a pure scale difference vanishes under normalisation, an injected one-sided tail survives. `--gate` and `--require-id` expose two asymmetries in the shipped derivation (anchor gates on gen and conditions on DeepTau; Delphes gates on reco and conditions on nothing).
8. **Map accuracy.** No map is statistics-limited at 2×10^5 anchor events (b-tag 0.25–0.42%/bin, τ_h eff 0.43–1.26%, escale 0.2–0.3%, p_T^{miss} 0.16–0.21%); what limits a map is np, not n. Two defects do not improve with statistics: the mass quantile grid interpolates across the empty region between the $\pi^{\pm}$ atom and the ρ shoulder ($\sim 6\times$ over-populated, measured inert), and the 300 GeV flat extrapolation (untested).
9. **Generator copies.** Delphes writes the full Pythia history, CMS `GenPart` is pruned. Rule: anything that **counts** generator objects must collapse copies first; anything that **matches** geometrically need not.

Reproducing.

```
# --max-events is REQUIRED: derive_maps caps the anchor from the config but reads
# the Delphes side uncapped.
python -m delphes_pipeline.tuning.derive_maps --config config.v1.yml \
    --output cards/tuning/maps_v1.json --max-events 200000
python -m delphes_pipeline.validation.run_validation --config config.v1.yml
python scripts/tauh_cutflow.py --config config.v1.yml [...] --gen-dr plots/gen_dr
python scripts/nsbi_overlay.py --config config.v1.yml [...] --tautau-only \
    --diagnostics --split-gen-matched
python scripts/tau_response.py --config config.v1.yml [...] --require-id
```